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(54) **SERVICE AWARE PROXY-CALL SESSION CONTROL FUNCTION (P-CSCF) TRAFFIC ROUTING**

(71) Applicant: **T-Mobile USA, Inc.**, Bellevue, WA (US)

(72) Inventors: **Khurram Ahmad Mirza**, Bellevue, WA (US); **Shujaur Mufti**, Snoqualmie, WA (US)

(73) Assignee: **T-Mobile USA, Inc.**, Bellevue, WA (US)

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CPC **H04L 12/1407** (2013.01); **H04L 47/12** (2013.01); **H04L 65/1016** (2013.01); **H04L 65/1045** (2022.05)

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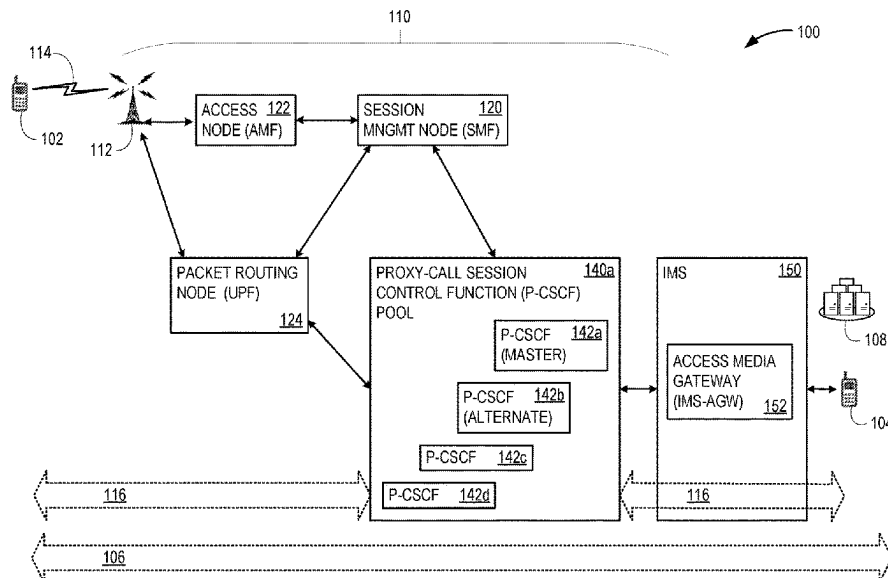
Primary Examiner — Syed Ali

(74) Attorney, Agent, or Firm — Foley IP Law, PLLC

(57) **ABSTRACT**

Solutions for service aware proxy-call session control function (P-CSCF) traffic routing include: receiving, at a master P-CSCF in a pool of a plurality of P-CSCFs, internet protocol (IP) multimedia subsystem (IMS) traffic from a device; determining, by the master P-CSCF, a traffic type (e.g., 5G, 4G, SMS, WiFi, etc.) of the IMS traffic from the device; based on at least the traffic type of the IMS traffic from the device, selecting, by the master P-CSCF, a serving P-CSCF from the pool for the device; and forwarding, by the serving P-CSCF for the device, the IMS traffic from the device to an IMS. In some examples, the serving P-CSCF selection also includes loading considerations. In some examples, an alternate P-CSCF is available for failover as the new master P-CSCF. In some examples, a network includes multiple pools of P-CSCFs, with each pool having its own master P-CSCF and alternate P-CSCF.

13 Claims, 6 Drawing Sheets



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FIG. 1

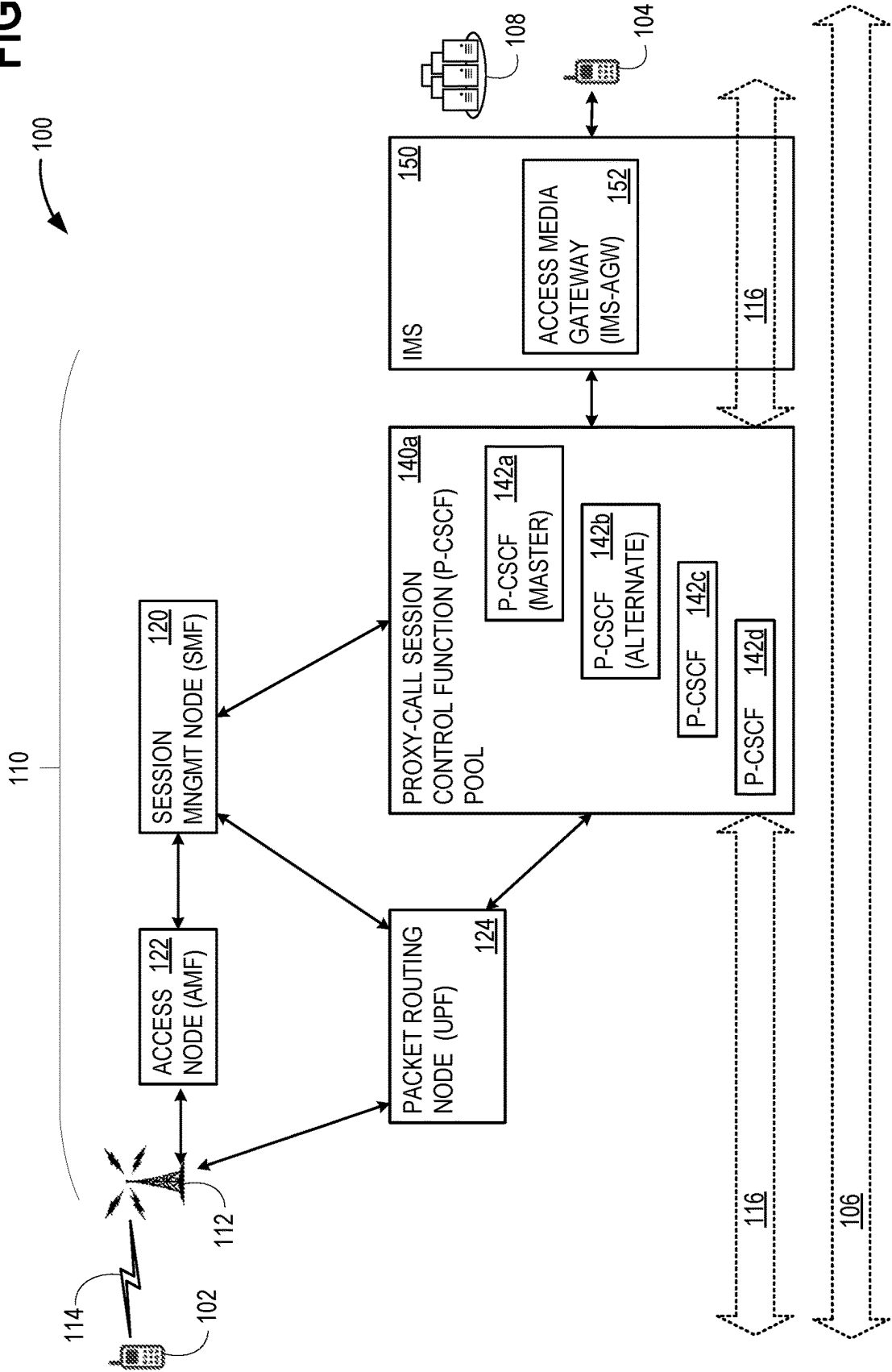
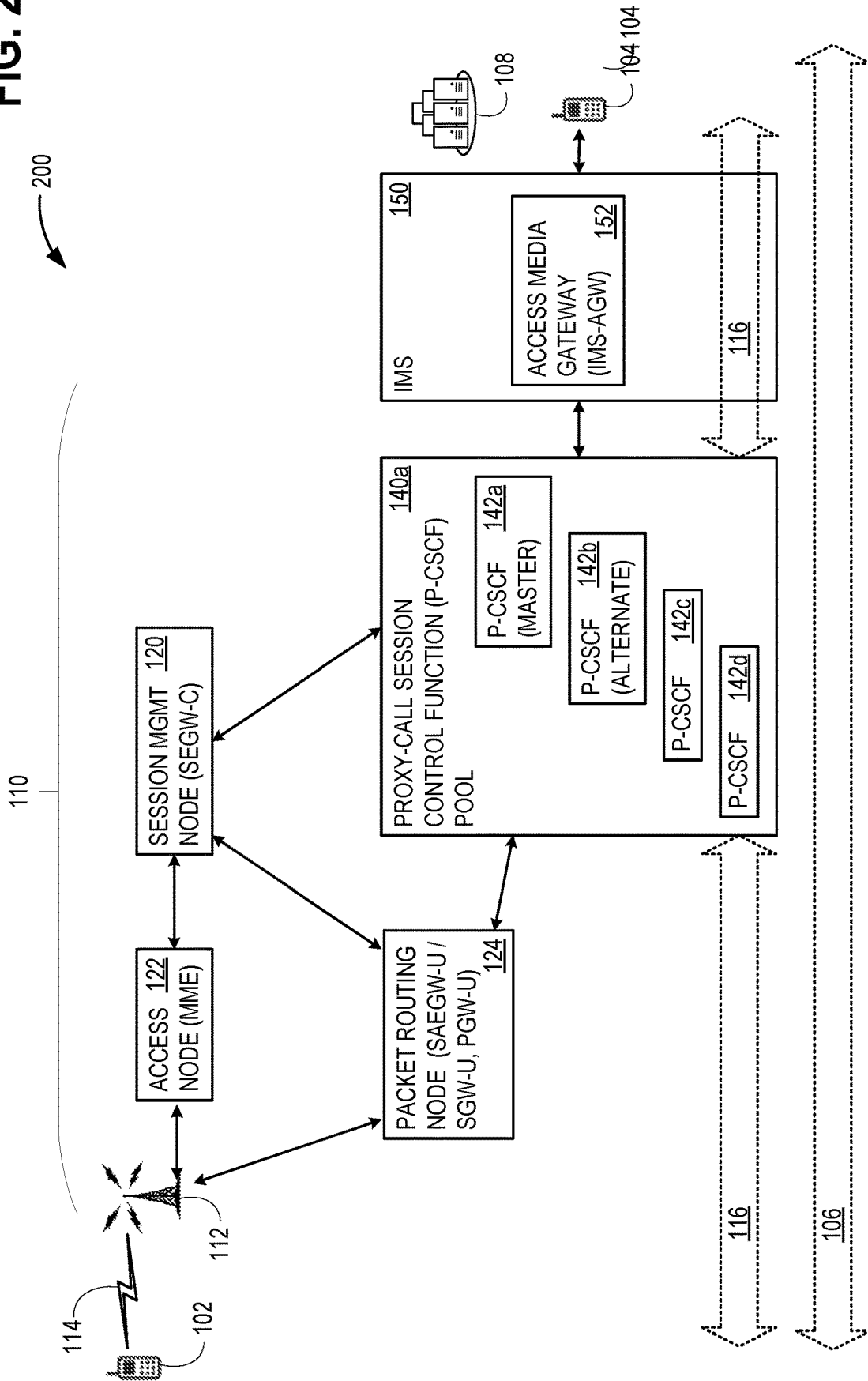


FIG. 2



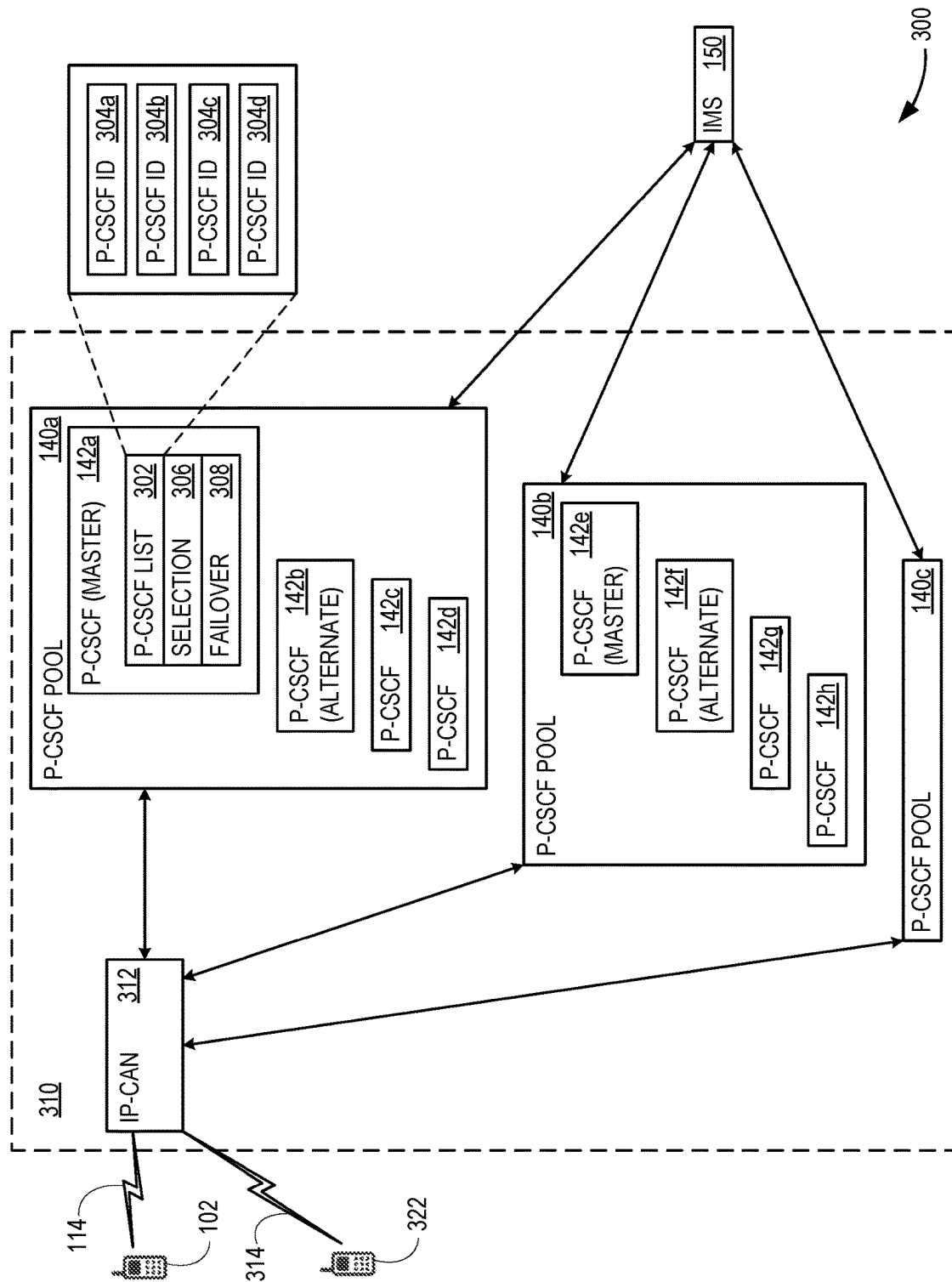
**FIG. 3**

FIG. 4

400

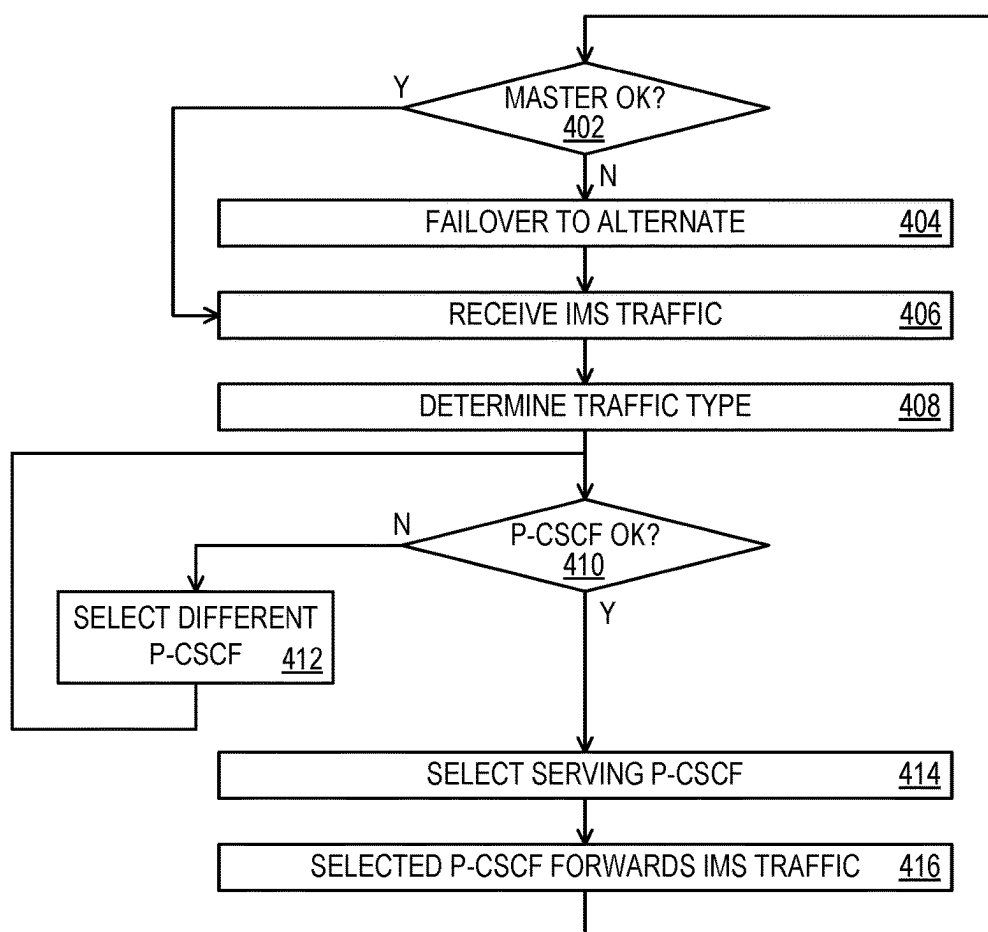


FIG. 5

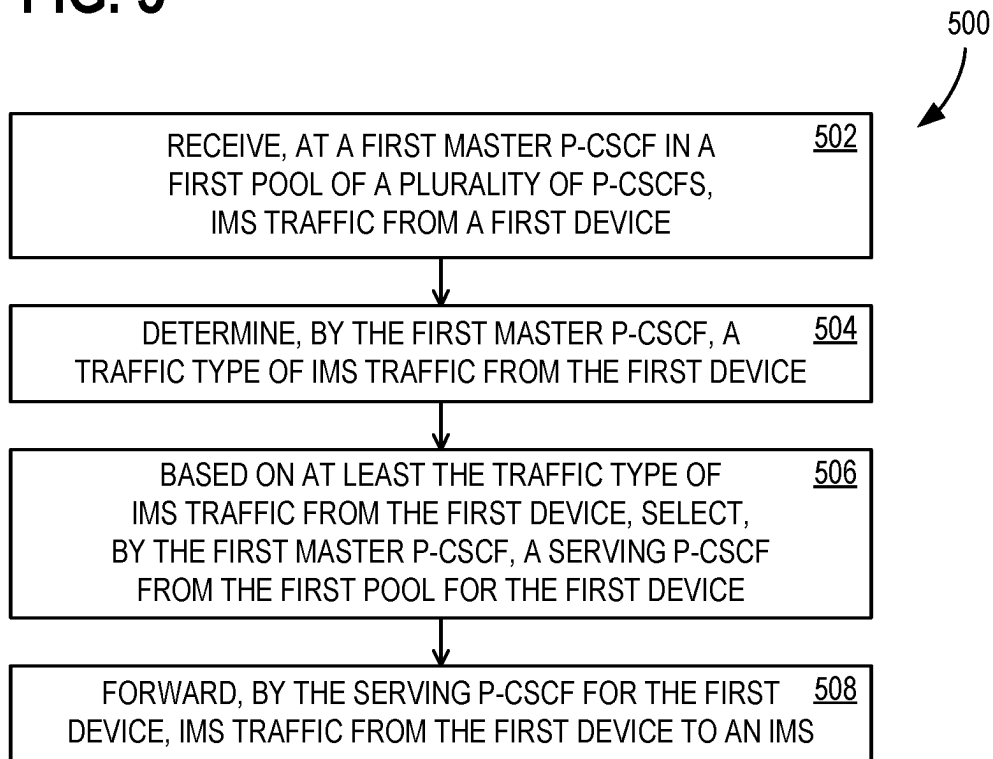
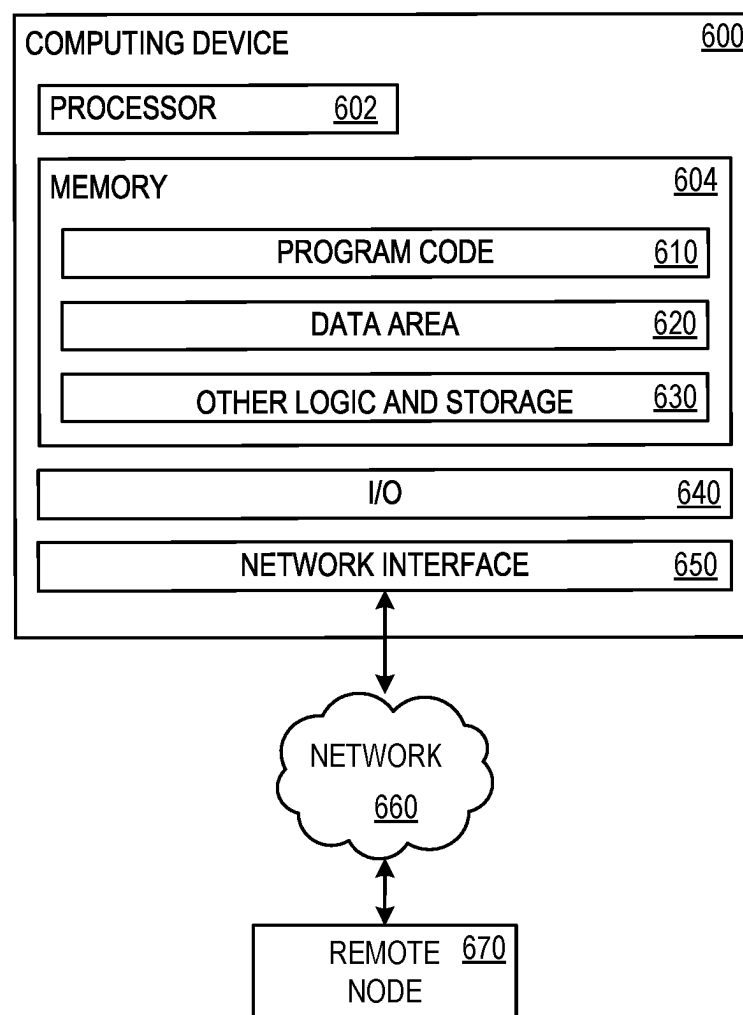


FIG. 6



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SERVICE AWARE PROXY-CALL SESSION CONTROL FUNCTION (P-CSCF) TRAFFIC ROUTING

BACKGROUND

A proxy-call session control function (P-CSCF) is a session initiation protocol (SIP) proxy that is the first point of contact for a user equipment (UE) in a mobile network. SIP traffic to and from a UE passes through a P-CSCF and an internet protocol (IP) multimedia subsystem (IMS). For example, in order to make a voice or video call, a UE sends a SIP message (e.g., a SIP invite) to a P-CSCF, which forwards it to the IMS, in order to reach the called party. Different types of IMS traffic, such as fifth generation (5G) cellular, fourth generation (4G) cellular, short message service (SMS), WiFi, in-vehicle local area network (LAN), and rich communication services (RCS), currently exist and must be supported by all P-CSCFs. When a P-CSCF ceases operation (e.g., due to being overloaded, maintenance, or some other reason) large numbers of users (e.g., humans with a UE, internet of things (IoT) devices, or others) lose service.

SUMMARY

The following summary is provided to illustrate examples disclosed herein, but is not meant to limit all examples to any particular configuration or sequence of operations.

Solutions for service aware proxy-call session control function (P-CSCF) traffic routing include: receiving, at a master P-CSCF in a pool of a plurality of P-CSCFs, internet protocol (IP) multimedia subsystem (IMS) traffic from a device; determining, by the master P-CSCF, a traffic type (e.g., 5G, 4G, SMS, WiFi, etc.) of the IMS traffic from the device; based on at least the traffic type of the IMS traffic from the device, selecting, by the master P-CSCF, a serving P-CSCF from the pool for the device; and forwarding, by the serving P-CSCF for the device, the IMS traffic from the device to an IMS. In some examples, the serving P-CSCF selection also includes loading considerations. In some examples, an alternate P-CSCF is available for failover as the new master P-CSCF. In some examples, a network includes multiple pools of P-CSCFs, with each pool having its own master P-CSCF and alternate P-CSCF. In some examples, the traffic type may be: fifth generation (5G) cellular, fourth generation (4G) cellular, short message service (SMS), WiFi, in-vehicle local area network (LAN), rich communication services (RCS), or another cellular generation.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosed examples are described below with reference to the accompanying drawing figures listed below, wherein:

FIG. 1 illustrates an exemplary arrangement that advantageously enables service aware proxy-call session control function (P-CSCF) traffic routing;

FIG. 2 illustrates an alternative configuration that also advantageously enables service aware P-CSCF traffic routing;

FIG. 3 illustrates a generic architecture that incorporates features of the arrangements of FIG. 1 and/or FIG. 2;

FIG. 4 illustrates a flowchart of exemplary operations that may be associated with examples of the arrangement of FIG. 3;

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FIG. 5 illustrates another flowchart of exemplary operations that may be associated with examples of the arrangement of FIG. 3; and

FIG. 6 illustrates a block diagram of a computing device suitable for implementing various aspects of the disclosure.

Corresponding reference characters indicate corresponding parts throughout the drawings. References made throughout this disclosure relating to specific examples, are provided for illustrative purposes, and are not meant to limit all implementations or to be interpreted as excluding the existence of additional implementations that also incorporate the recited features.

DETAILED DESCRIPTION

Solutions for service aware proxy-call session control function (P-CSCF) traffic routing include: receiving, at a master P-CSCF in a pool of a plurality of P-CSCFs, internet protocol (IP) multimedia subsystem (IMS) traffic from a device (e.g., a user equipment (UE) or another device); determining, by the master P-CSCF, a traffic type of the IMS traffic from the device; based on at least the traffic type of the IMS traffic from the device, selecting, by the master P-CSCF, a serving P-CSCF from the pool for the device; and forwarding, by the serving P-CSCF for the device, the IMS traffic from the device to an IMS. In some examples, the serving P-CSCF selection also includes loading considerations. In some examples, an alternate P-CSCF is available for failover as the new master P-CSCF. In some examples, a network includes multiple pools of P-CSCFs, with each pool having its own master P-CSCF and alternate P-CSCF. In some examples, the traffic type may be: fifth generation (5G) cellular, fourth generation (4G) cellular, short message service (SMS), WiFi, in-vehicle local area network (LAN) such as in-flight wireless, rich communication services (RCS), or another cellular generation such as third generation (3G) or (when it arrives) sixth generation (6G).

Aspects of the disclosure simplify and improve the operation of computer networks by permitting consolidation of traffic type on different P-CSCFs (e.g., providing a network slicing functionality), thereby relaxing the requirement that every P-CSCF support every type of traffic. Not only may this reduce the cost of provisioning P-CSCFs, but also, in some scenarios, a P-CSCF may exhibit superior performance by handling only limited traffic types. These advantages are enabled by a master P-CSCF selecting a serving P-CSCF for a device, based on at least the traffic type of the IMS traffic from the device. The master P-CSCF serves as a traffic sorter and load balancer for P-CSCFs in a pool (e.g., a region). The master P-CSCF distributes traffic to other P-CSCFs on the basis of their services and provides load balancing properties, fault/outage detection, distributed denial of service (DDOS) response, network robustness and resilience.

For example, a common core network may have six to ten pools of approximately a dozen or more P-CSCFs each. In a pool, two may cater to 5G traffic; four may cater to 4G traffic; one each may cater to SMS, RCS traffic, or in-flight wireless, or serve integration access; and two may handle any traffic type, as necessary. Two P-CSCFs will perform additional duties, one as the master P-CSCF, to act as an initial traffic entry point and steer traffic to one of the P-CSCFs in the pool (based on type and/or for load balancing), and one as an alternate master. As traffic types change (e.g., users migrating from 4G to 5G, the balance of P-CSCFs dedicated to 5G and 4G traffic may change. As 6G is rolled out, further significant advantageous savings may

be realized by reducing the number of P-CSCFs (if 6G uses P-CSCFs) that must be compatible with 6G, immediately upon 6G availability.

FIG. 1 illustrates an exemplary arrangement 100 that advantageously enables service aware P-CSCF traffic routing, for example when establishing a data traffic session 106 for a device 102 on a network 110. Data traffic sessions include voice and video calls between UEs and streaming video. In some examples, device 102 comprises a UE, and data traffic session 106 comprises a voice or video call with another UE 104. In some examples, device 102 streams video over data traffic session 106 from a remote resource 108. In some examples, device 102 comprises an internet of things (IoT) device, such as a connected appliance, a smart home security system, an autonomous factory or farming machine, a wearable health monitor, a wireless inventory tracker, or another device, which uploads or downloads data from remote resource 108 using data traffic session 106. Device 102 sets up data traffic session 106 using session initiation protocol (SIP) signaling 116, such as a SIP invite. SIP signaling 116 and data traffic session 106 are each IMS traffic, because they pass through IMS 150.

Data traffic session 106 passes from device 102, over an air interface 114, through a radio access network (RAN) 112, a packet routing node 124, a selected one of a plurality of P-CSCFs 142a-142d (in a P-CSCF pool 140a), and an IMS Access Media Gateway (IMS-AGW) 152 in an IMS 150. P-CSCF pool 140a comprises a pool of a plurality of P-CSCFs. Although only a single packet routing node 124 is shown, it should be understood that network 110 may have multiple packet routing nodes. In some examples, device 102 is assigned to packet routing node 124 via a session management node 120, and device 102 communicates with session management node 120 via an access node 122.

Arrangement 100 is shown as 5G cellular, in which packet routing node 124 comprises a user plane function (UPF), session management node 120 comprises a session management function (SMF), and access node 122 comprises an access and mobility management function (AMF). FIG. 2 illustrates an alternative 4G cellular configuration 200, in which access node 122 comprises a mobility management entity (MME), session management node 120 comprises a system architecture (SAE) evolution gateway—control plane (SAEGW-C), and packet routing node 124 comprises an SAE evolution gateway—user plane (SAEGW-U). An SAEGW-C is the combination of a serving gateway (SGW)—control plane (SGW-C) and a packet data network gateway (PGW)—control plane (PGW-C). An SAEGW-U is the combination of an SGW—user plane (SGW-U) and a PGW—user plane (PGW-U). The functional aspects of arrangement 200 are similar to those of arrangement 100, with respect to the operation of P-CSCF pool 140a.

FIG. 3 illustrates an arrangement 300 that incorporates features of the arrangements of FIG. 1 and/or FIG. 2. The various 5G and/or 4G components of network 110 (of FIGS. 1 and 2) are represented as an IP connectivity access network (IP-CAN) 312 within a common core network 310. Thus, device 102 communicates with an IMS 150 via air interface 114, IP-CAN 312, and P-CSCF pool 140a. A second device 322 communicates with an IMS 150 via air interface 314, IP-CAN 312, and a P-CSCF pool 140b. As indicated, common core network 310 comprises a plurality of P-CSCF pools: P-CSCF pool 140a, P-CSCF pool 140b, and a P-CSCF pool 140c. In some examples, P-CSCF pools 140a-140c are defined, at least in part, by geographical regions (e.g. devices 102 and 322 use different P-CSCF pools

because they are in different geographical regions). Each of P-CSCF pools 140a-140c comprises a plurality of P-CSCFs.

For example, as already indicates in FIGS. 1 and 2, P-CSCF pool 140a comprises P-CSCF 142a, P-CSCF 142b, P-CSCF 142c, and P-CSCF 142d. P-CSCF 142a performs as a master node for P-CSCF pool 140a, and P-CSCF 142b is an alternate master node in the event of a failover (e.g., if P-CSCF 142a is not functioning as a master node because it has ceased operation or has another problem). P-CSCF pool 140b is similarly structured, and comprises a P-CSCF 142e, a P-CSCF 142f, a P-CSCF 142g, and a P-CSCF 142h. P-CSCF 142e performs as a master node for P-CSCF pool 140b, and P-CSCF 142f is an alternate master node in the event of a failover (e.g., if P-CSCF 142e is not functioning as a master node because it has ceased operation or has another problem). P-CSCF pool 140c is also similarly configured.

P-CSCF 142a has a P-CSCF list 302 that identifies P-CSCFs within P-CSCF pool 140a, and their ability to handle IMS traffic, such as traffic loading and status (functioning or out of service). For example, P-CSCF identification (ID) 304a indicates loading and status for P-CSCF 142a, P-CSCF ID 304b indicates loading and status for P-CSCF 142b, P-CSCF ID 304c indicates loading and status for P-CSCF 142c, and P-CSCF ID 304d indicates loading and status for P-CSCF 142d. A selection logic 306 selects a serving P-CSCF (e.g., one of P-CSCFs 142a-142d) for device 102 based on at least the traffic type and using loading and status information in P-CSCF list 302.

P-CSCF 142a is also shown as including a failover control 308, although failover control 308 may be located within IP-CAN 312 (e.g., within session management node 120), within P-CSCF 142b, and/or within another node within P-CSCF pool 140a. In the event that P-CSCF 142a is unable to function properly as the master node for P-CSCF pool 140a, failover control 308 performs a failover to P-CSCF 142b so that P-CSCF 142b functions as the master node for P-CSCF pool 140a. P-CSCF 142e and the master P-CSCF for P-CSCF pool 140c are configured similarly P-CSCF 142a, and P-CSCF pools 140b and 140c have similar failover capability. The operations of arrangement 300 with respect to service aware P-CSCF traffic routing are described in relation to FIG. 4.

FIG. 4 illustrates a flowchart 400 of exemplary operations that may be associated with examples of arrangement 300 (and thus, by extension, also arrangements 100 and 200). That is, flowchart 400 identifies exemplary operations associated with service aware P-CSCF traffic routing. In some examples, at least a portion of flowchart 400 may each be performed using one or more computing devices 600 of FIG. 6. A decision operation 402 determines whether (initially-designated) master P-CSCF 142a in P-CSCF pool 140a (e.g., a pool of a plurality of P-CSCFs) is functioning as a master node for P-CSCF pool 140a. If it is not, then operation 404 includes, based on at least determining that P-CSCF 142a is not functioning as the master node for P-CSCF pool 140a, performing a failover to P-CSCF 142b (the alternate master P-CSCF) in P-CSCF pool 140a. For the remainder of flowchart 400 (until P-CSCF 142a is restored as the master node, alternate master P-CSCF 142b functions as the master node for P-CSCF pool 140a.

Operation 406 includes receiving, at P-CSCF 142a (if P-CSCF 142a is performing as the master P-CSCF, or P-CSCF 142b, if P-CSCF 142b is performing as the master P-CSCF), IMS traffic from device 102. In some examples, device 102 comprises a cellular UE mobile device or an IoT device. In some examples, the IMS traffic comprises a

session SIP invite. Operation **408** includes determining, by master P-CSCF **142a**, a traffic type of the IMS traffic from device **102**. In some examples, the traffic type is selected from a list consisting of: 5G cellular, 4G cellular, SMS, WiFi, in-vehicle LAN (e.g., in-flight WiFi), RCS, and another cellular generation.

Decision operation **410** includes determining, by master P-CSCF **142a**, whether a P-CSCF that is a candidate for being the serving P-CSCF for device **102** is able to handle traffic for device **102**. In some examples, determining whether a P-CSCF is able to handle traffic for a device includes both determining whether the P-CSCF is properly functioning (e.g., is not down), and also determining whether the P-CSCF is overloaded (e.g., determining load balancing of a particular P-CSCF relative to other available P-CSCFs). If a P-CSCF is overloaded or down, it is not selected as the serving P-CSCF, for example by identifying it as not suitable at **412**, so that another P-CSCF will instead be considered for selection in decision operation **410**.

When a suitable P-CSCF is located, operation **414** includes, based on at least the traffic type of the IMS traffic from device **102**, selecting, by master P-CSCF **142a**, a serving P-CSCF for device **102**. In some example, load balancing is also a selection criteria, and selecting the serving P-CSCF for device **102** comprises: selecting the serving P-CSCF for device **102** based on at least the traffic type of the IMS traffic from device **102** and based on at least determining that the serving P-CSCF for device **102** is able to handle traffic for device **102**. For example, master P-CSCF **142a** may select itself or P-CSCF **142c**. However, if P-CSCF **142a** may select itself or P-CSCF **142c** are heavily loaded, and P-CSCF **142a** handles a different traffic type (e.g., 4G versus 5G), P-CSCF **142a** may instead select P-CSCF **142d**.

Operation **416** includes forwarding, by the serving P-CSCF for device **102**, the IMS traffic from device **102** to IMS **150**. In some examples, IMS **150** comprises IMS-AGW **152**. Flowchart **400** returns to decision operation **402** to determine whether a master node failover is needed.

Additional implementations of flowchart **400** operating in parallel for P-CSCF pools **140b** and **140c**. In a parallel version of flowchart **400** for P-CSCF pool **140b**, for example, operation **406** includes receiving, at P-CSCF **142e** (a second master P-CSCF) in P-CSCF pool **140b** (a second pool of a plurality of P-CSCFs), IMS traffic from device **322**, wherein P-CSCF pool **140a** and P-CSCFs pool **140b** reside within common core network **310**. Operation **408** in the parallel flowchart **400** includes determining, by P-CSCF **142e**, a traffic type of the IMS traffic from device **322**. Operation **414** in the parallel flowchart **400** includes, based on at least the traffic type of the IMS traffic from device **322**, selecting, by P-CSCF **142e**, a serving P-CSCF for device **322**. Operation **416** in the parallel flowchart **400** includes forwarding, by the serving P-CSCF for device **322**, the IMS traffic from device **322** to IMS **150**.

FIG. **5** illustrates a flowchart **500** of exemplary operations associated with service aware P-CSCF traffic routing. In some examples, at least a portion of flowchart **500** may be performed using one or more computing devices **600** of FIG. **6**. Operation **502** includes receiving, at a first master P-CSCF in a first pool of a plurality of P-CSCFs, IMS traffic from a first device. Operation **504** includes determining, by the first master P-CSCF, a traffic type of the IMS traffic from the first device. Operation **506** includes based on at least the traffic type of the IMS traffic from the first device, selecting, by the first master P-CSCF, a serving P-CSCF from the first pool for the first device. Operation **508** includes forwarding,

by the serving P-CSCF for the first device, the IMS traffic from the first device to an IMS.

FIG. **6** illustrates a block diagram of computing device **600** that may be used as a component of arrangement **100**, for example, as any component described herein that may require computational or storage capacity. Computing device **600** has at least a processor **602** and a memory **604** that holds program code **610**, data area **620**, and other logic and storage **630**. Memory **604** is any device allowing information, such as computer executable instructions and/or other data, to be stored and retrieved. For example, memory **604** may include one or more random access memory (RAM) modules, flash memory modules, hard disks, solid-state disks, persistent memory devices, and/or optical disks. Program code **610** comprises computer executable instructions and computer executable components including any instructions necessary to perform operations described herein. Data area **620** holds any data necessary to perform operations described herein. Memory **604** also includes other logic and storage **630** that performs or facilitates other functions disclosed herein or otherwise required of computing device **600**. An input/output (I/O) component **640** facilitates receiving input from users and other devices and generating displays for users and outputs for other devices. A network interface **650** permits communication over a network **660** with a remote node **670**, which may represent another implementation of computing device **600**.

Additional Examples

An example method of service aware P-CSCF traffic routing comprises: receiving, at a first master P-CSCF in a first pool of a plurality of P-CSCFs, IMS traffic from a first device; determining, by the first master P-CSCF, a traffic type of the IMS traffic from the first device; based on at least the traffic type of the IMS traffic from the first device, selecting, by the first master P-CSCF, a serving P-CSCF from the first pool for the first device; and forwarding, by the serving P-CSCF for the first device, the IMS traffic from the first device to an IMS.

An example system for service aware P-CSCF traffic routing comprises: a processor; and a computer-readable medium storing instructions that are operative upon execution by the processor to: receive, at a first master P-CSCF in a first pool of a plurality of P-CSCFs, IMS traffic from a first device; determine, by the first master P-CSCF, a traffic type of the IMS traffic from the first device; based on at least the traffic type of the IMS traffic from the first device, select, by the first master P-CSCF, a serving P-CSCF from the first pool for the first device; and forward, by the serving P-CSCF for the first device, the IMS traffic from the first device to an IMS.

One or more example computer storage devices has computer-executable instructions stored thereon, which, upon execution by a computer, cause the computer to perform operations comprising: receiving, at a first master P-CSCF in a first pool of a plurality of P-CSCFs, IMS traffic from a first device; determining, by the first master P-CSCF, a traffic type of the IMS traffic from the first device; based on at least the traffic type of the IMS traffic from the first device, selecting, by the first master P-CSCF, a serving P-CSCF from the first pool for the first device; and forwarding, by the serving P-CSCF for the first device, the IMS traffic from the first device to an IMS.

Alternatively, or in addition to the other examples described herein, examples include any combination of the following:

determining, by the first master P-CSCF, whether the serving P-CSCF for the first device is able to handle traffic for the first device, wherein selecting the serving P-CSCF for the first device comprises: selecting the serving P-CSCF for the first device based on at least the traffic type of the IMS traffic from the first device and based on at least determining that the serving P-CSCF for the first device is able to handle traffic for the first device;

determining whether a P-CSCF is able to handle traffic for a device includes both determining whether the P-CSCF is properly functioning, and also determining whether the P-CSCF is overloaded;

determining whether the first master P-CSCF in the first pool of a plurality of P-CSCFs is functioning as a master node for the first pool of a plurality of P-CSCFs; based on at least determining that the first master P-CSCF in the first pool of a plurality of P-CSCFs is not functioning as the master node for the first pool of a plurality of P-CSCFs, performing a failover to an alternate master P-CSCF in the first pool of a plurality of P-CSCFs;

functioning, by the alternate master P-CSCF, as the master node for the first pool of a plurality of P-CSCFs;

receiving, at a second master P-CSCF in a second pool of a plurality of P-CSCFs, IMS traffic from a second device, wherein the first pool of a plurality of P-CSCFs and the second pool of a plurality of P-CSCFs reside within a common core network;

determining, by the second master P-CSCF, a traffic type of the IMS traffic from the second device;

based on at least the traffic type of the IMS traffic from the second device, selecting, by the second master P-CSCF, a serving P-CSCF from the second pool for the second device;

forwarding, by the serving P-CSCF for the second device, the IMS traffic from the second device to the IMS;

the traffic type is selected from a list consisting of: 5G cellular, 4G cellular, SMS, WiFi, in-vehicle LAN, RCS, and another cellular generation;

the first device comprises a cellular UE mobile device or an IoT device;

the IMS comprises an IMS-AGW; and

the IMS traffic comprises a SIP invite.

The order of execution or performance of the operations in examples of the disclosure illustrated and described herein is not essential, unless otherwise specified. That is, the operations may be performed in any order, unless otherwise specified, and examples of the disclosure may include additional or fewer operations than those disclosed herein. For example, it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the disclosure. It will be understood that the benefits and advantages described above may relate to one embodiment or may relate to several embodiments. When introducing elements of aspects of the disclosure or the examples thereof, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The term “exemplary” is intended to mean “an example of.”

Having described aspects of the disclosure in detail, it will be apparent that modifications and variations are possible without departing from the scope of aspects of the disclosure as defined in the appended claims. As various changes may be made in the above constructions, products, and methods without departing from the scope of aspects of the disclosure, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

What is claimed is:

1. A method of service aware proxy-call session control function (P-CSCF) traffic routing, the method comprising: receiving, at a master P-CSCF, a session initiation protocol (SIP) invite for internet protocol (IP) multimedia subsystem (IMS) traffic from a mobile device that is a fifth generation (5G) capable device;
- determining, by the master P-CSCF, a specific traffic type of the IMS traffic prior to serving P-CSCF assignment, the specific traffic type of the IMS traffic being identified from a list including a 5G cellular traffic type, a Short Messaging Service (SMS) traffic type, a Wi-Fi traffic type, and an in-vehicle local area network (LAN) traffic type;
- based on at least the specific traffic type of the IMS traffic, selecting, by the master P-CSCF, a serving P-CSCF to assign to the mobile device based at least partially on the serving P-CSCF supporting the specific traffic type of the IMS traffic; and
- forwarding, by the master P-CSCF, the SIP invite to the serving P-CSCF.
2. The method of claim 1, wherein the mobile device is a cellular user equipment (UE).
3. The method of claim 1, wherein the mobile device is an internet of things (IoT) device.
4. The method of claim 1, wherein the specific traffic type of the IMS traffic is the 5G cellular traffic type.
5. The method of claim 1, wherein the specific traffic type of the IMS traffic is the SMS traffic type.
6. The method of claim 1, wherein the specific traffic type of the IMS traffic is the Wi-Fi traffic type.
7. The method of claim 1, wherein the specific traffic type of the IMS traffic is the in-vehicle LAN traffic type.
8. A master proxy-call session control function (P-CSCF) comprising:
 - a processor; and
 - computer-readable medium storing programming instructions that, when executed by the processor, cause the processor to:
 - receive a session initiation protocol (SIP) invite for internet protocol (IP) multimedia subsystem (IMS) traffic from a mobile device that is a fifth generation (5G) capable device;
 - determine a specific traffic type of the IMS traffic received from the mobile device prior to serving P-CSCF assignment, the specific traffic type of the IMS traffic being identified from a list including a 5G cellular traffic type, a Short Messaging Service (SMS) traffic type, a Wi-Fi traffic type, and an in-vehicle local area network (LAN) traffic type;
 - based on at least the specific traffic type of the IMS traffic, select a serving P-CSCF to assign to the mobile device based at least partially on the serving P-CSCF supporting the specific traffic type of the IMS traffic; and
 - forward the SIP invite to the serving P-CSCF.
9. The master P-CSCF of claim 8, wherein the mobile device is a cellular user equipment (UE).

10. The master P-CSCF of claim 8, wherein the mobile device is an internet of things (IoT) device.

11. One or more computer storage devices storing programming instructions that, when executed by a processor, cause the processor to:

receive, at a master P-CSCF, a session initiation protocol (SIP) invite for internet protocol (IP) multimedia subsystem (IMS) traffic from a mobile device that is a fifth generation (5G) capable device;

determine, by the master P-CSCF, a specific traffic type of the IMS traffic prior to serving P-CSCF assignment, the specific traffic type of the IMS traffic being identified from a list including a 5G cellular traffic type, a Short Messaging Service (SMS) traffic type, a Wi-Fi traffic type, and an in-vehicle local area network (LAN) traffic type;

based on at least the specific traffic type of the IMS traffic, select, by the master P-CSCF, a serving P-CSCF to assign to the mobile device based at least partially on the serving P-CSCF supporting the specific traffic type of the IMS traffic; and

forward, by the master P-CSCF, the SIP invite to the serving P-CSCF.

12. The one or more computer storage devices of claim 11, wherein the mobile device is a cellular user equipment (UE).

13. The one or more computer storage devices of claim 11, wherein the mobile device is an internet of things (IoT) device.

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